FIN 400 - Analytical Methods in Finance - Research Project One

Using your chosen dataset, this assignment will help you to understand your two variables and perform an initial investigation into how they relate to each other. This report will provide the foundation for the regression analysis your team will perform in Project Two.

The deliverable is an 8-10 page pdf submitted on Learning Suite. This limit includes all tables, figures, references, and any title page (should you choose to create one). Organize the report using Sections 1-6 below and label your answers to each part (this framework will ease grading for the TAs). Tables and figures need descriptive titles, units, and readable labels. Discuss the findings in the text. Professional presentation is a fairly significant part of the grade. This write-up assumes that your X and Y variables are quantitative and have some meaningful amount of variation. If either of the variables is categorical or a constant, check with me about appropriate ways to adjust.

1. **Research question and data**
    a. State your research question clearly. Identify one of your variables as your "Y" variable, which is the outcome you want to understand, and your "X" variable, which is the variable you think may explain differences in Y in some interesting way. Explain why you assigned these variables to these roles. What relationship were you expecting to find before the below analysis? Why?
    b. Define both of the variables clearly (including their units). Explain what one row of data represents, such as a single company in a single month or a single player in a single game. State the dates, people, firms, places, or other units covered by the data as applicable. X and Y in a row must refer to the same observation, with any intended timing difference explained.
    c. Identify and cite the sources for your data. Explain how you found those sources and how you used the research librarian (Maggie Marchant), if applicable. Describe anything you did to transform your data such as calculations, unit conversions, or combining of datasets in order to generate X and Y. These transformations could include things like converting your variables to percentage changes, logs, or per capita figures. These types of transformations are encouraged in order to align the data to your research question, but these transformed variables should be your X and Y variables throughout the project.
    d. Describe any data issues you encountered such as missing values, duplicate records, and apparent errors, and detail how you dealt with them. Use one set of rows with valid X and Y values for all parts of the report. Report the original number of rows in the data as downloaded, the number of rows excluded, and the final number used. Always keep a copy of the original data.
    e. What population or data generating process are you learning about in this project? Explain whether your data include that entire population or is just a sample from it, and identify one limitation on applying your findings more broadly (i.e., if it's a sample, what is one way in which it may not extrapolate or generalize to interesting, broader, settings).
2. **Describe each variable**
    a. Create one summary statistics table with a column for X and a column for Y. Report the observation count, mean, standard deviation, variance, minimum, maximum, and the 10th, 25th, 30th, 50th, 70th, 75th, and 90th percentiles. Use Excel's inclusive method for percentiles.
    b. State whether you used sample or population formulas for variance and standard deviation, and explain how that choice follows from Section 1e. Hold this convention fixed throughout the report.
    c. Create a histogram for each variable individually (or a frequency bar chart if it takes only a few distinct values). Describe its center, spread, and shape. Discuss any asymmetry, multiple modes, or unusually extreme values if present. Use both the graph and a mean-median comparison to support your interpretations.

3. Compare the observed data with a normal model

Complete parts a–c for both X and Y using the full dataset. For the normal-model calculations, treat the observed mean and standard deviations as the model's parameters. If the data are a sample, these are estimates of the population parameters. Calculating a z-score as if the data were normal does not require the data to actually be normally distributed.

a. Calculate the percentage of observations that are at or below the variable's mean. Interpret this percentage and explain how it relates to the shape of the distribution. Calculate the percentage by counting actual observations, (i.e., do not impose a normal model).
b. Calculate the z-score for the median observation. Assuming a normal distribution with the mean and standard deviation specified above, what percentage of values would be at or below your observed median? Explain why this answer differs from 50%, and under what conditions it would equal 50%.
c. Calculate the 90th percentile predicted by that same normal model. Compare it with the observed 90th percentile from Section 2. Considering both this comparison and the histogram, describe one way the normal model does or does not fit your data. Keep in mind that just because one particular percentile agrees with a normal distribution; you do not necessarily have a normally distributed variable.
d. Find the observation with the largest absolute z-score for Y. Identify that record and report its X and Y values and the z-score for both X and Y. Which variable is further from its own mean in standard-deviation units for that observation? Investigate whether this extreme Y value is plausible or appears to be an error. Do not remove a value simply because it is extreme or appears erroneous.

4. Investigate how Y varies with X

a. Create a scatterplot with X on the horizontal axis and Y on the vertical axis. Each point in the plot must represent exactly one row of your data. Describe whether higher X values tend to accompany higher Y values, lower Y values, or whether there appears to be no clear pattern. Discuss any curvature, clusters, or extreme observations that might be important.
b. Split the observations into two groups using the full dataset's median X: lower X includes values at or below the median, and higher X includes values above it. Keep tied values together; the groups do not necessarily need to contain an equal numbers of observations. Report the cutoff and each group's size. If either of the two groups has fewer than 10 observations, come talk to me so we can figure out an alternative cutoff or an appropriate adaptation.
c. For each group of X: report the mean, median, and standard deviation of Y. Calculate the higher-X group's mean Y minus the lower-X group's mean Y. Interpret that difference in Y's units. Does the comparison agree with the pattern in your scatterplot?
d. Compare the difference between the two group means with the dispersion of Y within each group. Does knowing whether X is high or low appear useful for predicting Y? How much uncertainty around Y remains even within groups? Support your answer with both the statistics you have calculated and your scatterplot.

Note that this process is a simple discretization as we did in class. Use the full dataset's cutoffs consistently. Do not recalculate a median separately within a group. A different cutoff may be used if it has a clear meaning for your research question: explain the choice and keep it fixed for both Sections 4 and 5.

5. Use conditional probabilities to study the relationship

For this section, imagine selecting one row at random from the dataset, with every row equally likely. Define event A as "X is in the higher-X group" from Section 4. Choose and explain a meaningful cutoff for Y, using the full dataset's median Y if no natural cutoff is available. Define B as "Y is above this cutoff." The complements A' and B' contain the lower-X and lower-Y observations. Place values equal to a cutoff into the lower group. Apply the same small-group guidance laid out in 4b to the Y split.

a. Create one 2 × 2 contingency table for the entire dataset using the layout below. Report counts and include row totals, column totals, and the overall total. Then report a second version in which every count is divided by the overall total (a probability, rather than a frequency, contingency table). The four interior percentages should sum to 100%. Use descriptive labels based on your actual variables not my placeholders.

	Higher Y (B)	Lower Y (B')	**Total**
Higher X (A)			
Lower X (A')			
Total			

b. Calculate $P(A \cap B)$ and $P(A \cup B)$. Interpret each using your variables.

c. Calculate $P(B)$, $P(B \mid A)$, and $P(B \mid A')$. Interpret each with a single sentence that identifies the relevant group. How does knowing whether X is high or low change the probability that Y is high? Report the difference between the two conditional probabilities in percentage points.

d. Calculate $P(A \mid B)$ using Bayes' rule and verify your answer directly from the counts. Explain how this question differs from asking for $P(B \mid A)$, even if the numerical answers happen to match.

e. Do the proportions in your table satisfy the independence condition for A and B? Interpret this.

f. Compare $P(A \cap B)$ and $P(A) \cdot P(B)$ using unrounded calculations, and relate the result to part c. Describe the size of the difference rather than reporting only whether it is zero (This check describes your dataset; it does not settle whether the relationship persists in a wider population. Independence of these particular high/low categories would not establish independence of X and Y at every possible value).

6. Answer your research question and plan the next step

a. Give your preliminary answer to your research question using at least two specific results and findings from your above report. Explain whether or not the findings match your original expectations. An unclear or unexpected relationship is perfectly acceptable.

b. Identify one additional variable you might like to add to your dataset that might help explain this X-Y relationship. Explain how it could relate to both X and Y, and how that could affect your interpretations. State what your current analysis leaves unresolved, including why any observed relationship alone does not establish that "X causes Y."